# BlackSpot Miner: A Case Study-Based Data Mining Analysis of Severe Road Collisions

From Data to Decisions: Problem → Data → Preprocessing → Algorithm → Results → Evaluation → Insights → Decision

| Group D15A_9 | Domain | Case data | Reference paper | Date |
|---|---|---|---|---|
| Jaiprakash Talreja (54)<br>Adarsh Singh (52)<br>Vidushi Singh (53)<br>Piyush Shukla (49) | Transportation and road safety | UK Department for Transport STATS19 collisions, 2021 to 2025 | Kumar and Toshniwal (2015), Journal of Big Data 2:26 | 1 November 2026 |

## 1. Problem and business objective

### 1.1 Problem statement

Road crashes kill about 1.19 million people a year worldwide [1]. India alone recorded about 4.8 lakh accidents and 1.72 lakh deaths in 2023 [2]. UN Sustainable Development Goal target 3.6 calls for these deaths and injuries to be halved [3].

A road authority cannot rebuild every junction. It can fund a limited number of sites each year, so the practical problem is a choice: which places, and what should engineers examine when they get there? The usual approaches fall short in three ways.

- **Ranking by last year's crash count.** A bad year is partly bad luck, so sites picked this way tend to improve the next year even if nothing is done. The list then seems to work when it has not.
- **Treating all crashes alike.** A road with many minor shunts may need less attention than one with fewer crashes that kill or seriously injure people.
- **Reading one condition at a time.** Risk often sits in combinations, such as a motorcycle on a 60 mph road, which a single-column summary cannot show.

### 1.2 Business objective

**Decision maker:** the road safety team of a highway authority. **Objective:** spend a fixed safety budget where it covers the most severe crashes. Severe here means killed or seriously injured, written KSI throughout this report. The analysis has three goals.

| Goal | Question | What the authority gets |
|---|---|---|
| **Where** | Which street-scale places concentrate KSI collisions? | A map of hotspots |
| **Why** | Which combinations of conditions raise the chance of KSI? | Tested rules that suggest what to review |
| **What next** | Which sites are likely to record KSI collisions next year? | A ranked priority list |

**Success test.** The share of next year's KSI collisions that fall inside the top N ranked sites, compared with a list built from past crash counts alone.

### 1.3 Why this case and this data

The reference paper by Kumar and Toshniwal [4] mines 11,574 road accidents from Dehradun, India. It is open access and uses the syllabus methods of clustering and association rules, but it has no predictive model, no coordinates and no test on later data. India's eDAR (formerly iRAD) system now captures GPS-tagged accidents across all 36 States and UTs [5], but we could not find a public record-level download. Great Britain publishes every police-reported injury collision with coordinates [6], so we use that data to build and test the method. The pipeline is designed to run on eDAR records in the same way. The numbers themselves do not transfer to India.

## 2. Data understanding and source

**Source:** Department for Transport, Road Safety Open Data (STATS19), latest five years, published under the Open Government Licence v3.0 [6]. Police record each injury collision on a national form. Each row is a real, reported event.

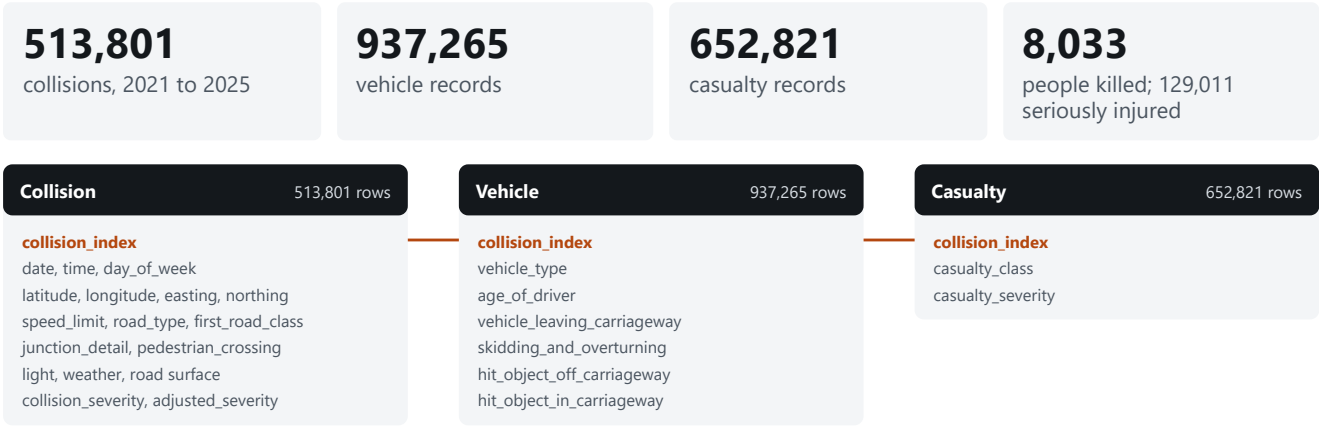



**Figure 1.** The three linked tables, joined on `collision_index`.

| Attribute group | Attribute type | Use in the analysis |
|---|---|---|
| Location: easting, northing, latitude, longitude | Ratio (metres, degrees) | Spatial distance for hotspot discovery |
| Road: speed limit, class, layout, junction, crossing | Ordinal and nominal | Rule items and model features |
| Conditions: lighting, weather, surface, time, day | Nominal, plus derived time bands | Rule items and model features |
| People and vehicles: type, driver age, pedestrian hit | Nominal, binary and ratio | Rule items and model features |
| Target: collision severity | Ordinal (Fatal, Serious, Slight) | Grouped into KSI against Slight |

### A data problem that changes the design

Police forces are moving to injury-based reporting. The officer records the injury and the severity is derived from it. DfT's own early analysis found this switch added between 5 and 15% to serious injuries [7]. In this data, the injury-based share of collisions rose from 49.6% in 2021 to 86.5% in 2025, and 15 of 44 police force codes changed system during the period. Raw KSI share therefore partly measures a change of form. DfT publishes a probability for each collision that restates it as if every force used the injury-based system [7], and this analysis uses it.

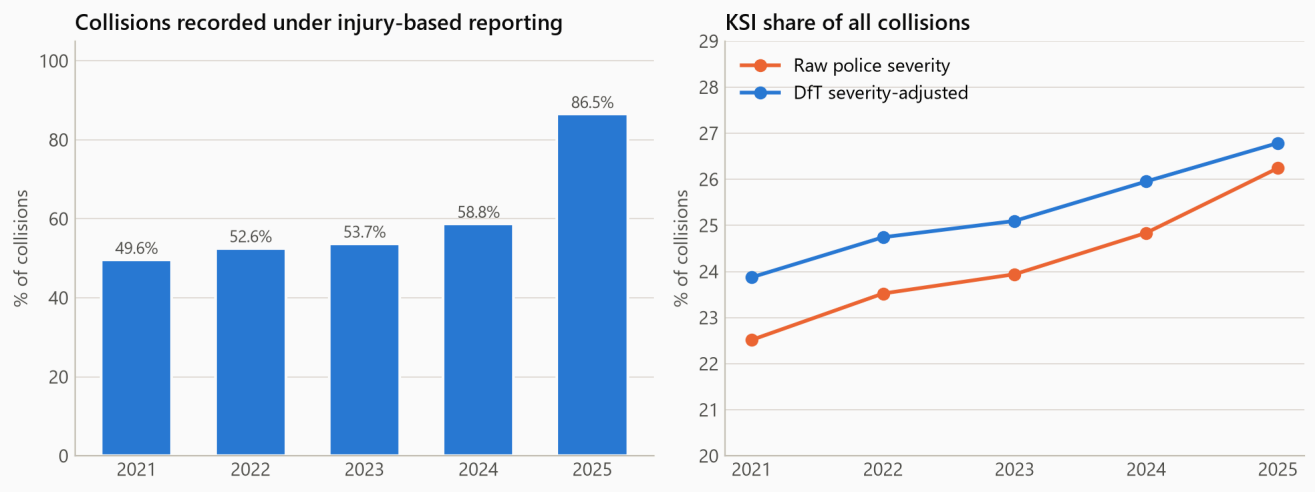



**Figure 2.** Reporting change. Raw KSI share rises from 22.51% to 26.24%. After adjustment the rise is smaller, 23.87% to 26.79%.

## 3. Data preprocessing (Module II)

Every field was audited year by year before use, because the 2024 STATS19 specification redefined several of them. Figure 3 shows the audit. Fields with an underlying coding change would teach a model the change of form, not road risk.

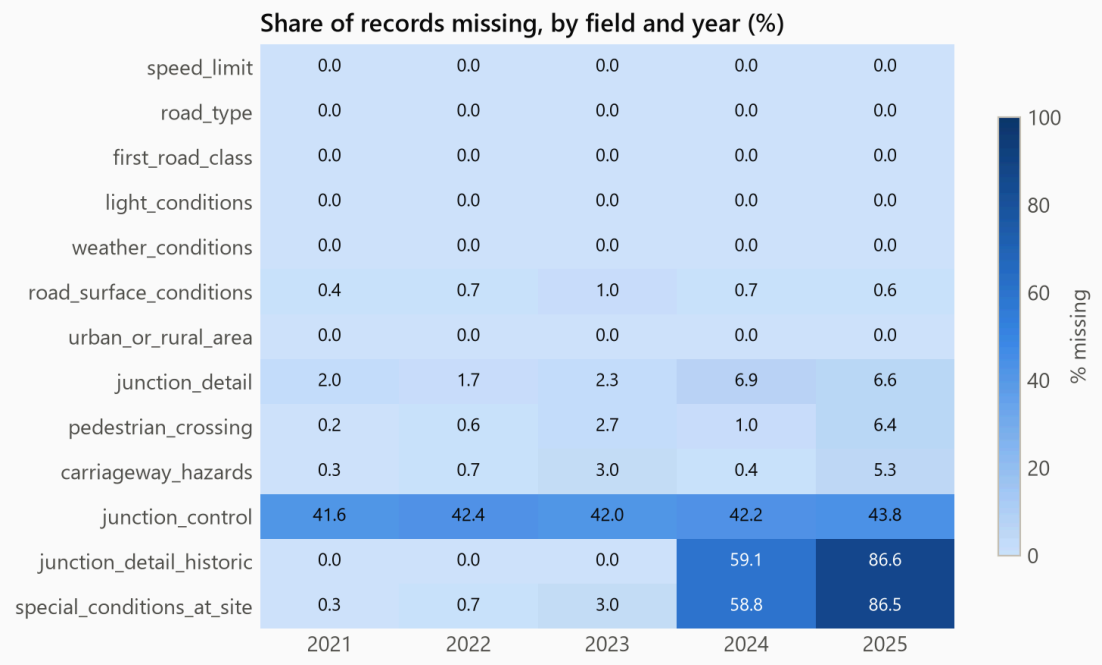



**Figure 3.** Field audit. `special_conditions_at_site` is 86.5% missing in 2025, so it is unusable even though earlier years look complete. The missing values for `junction_control` are structural: nearly all of them are collisions not at a junction, where the field does not apply.

| Step | What is done | Why |
|---|---|---|
| **Data integration** | Join vehicle and casualty tables to collisions; derive flags such as motorcycle involved, pedestrian hit, overturned | Who was involved is a stronger signal than road context |
| **Data cleaning** | Remove 56 records without a location or speed limit; map -1, 9 and 99 to an explicit Unknown | Unknown is not the same as a guessed value |
| **Attribute subset selection** | Drop `number_of_casualties`, police attendance and enhanced severity | They are known only after the outcome and would leak the label |

| | | |
|---|---|---|
| **Attribute subset selection** | Drop `*_historic` fields and `special_conditions_at_site` | Mostly empty in recent years |
| **Label consistency** | Rules and severity models use only injury-based records (309,041). Location ranking uses DfT adjusted KSI on all records (513,745) | A serious injury must mean the same thing in every row |
| **Data transformation** | Map integer codes to labels from the official guide; one-hot encode for classifiers; item encode for Apriori | Codes are nominal and have no numeric order |
| **Discretization** | Bin hours into five bands; group weather, surface and crossing types | Rare categories cannot support rules |
| **Aggregation** | Build 500 m grid cells with three-year history features | Budgets are spent on places, not on single crashes |
| **Split by time** | Train on 2021 to 2024, test on 2025 | A random split would let the model see the future |

## 4. Identifying the data mining tasks

| Business question | Data mining task | Type | Output | Module |
|---|---|---|---|---|
| Where do KSI collisions concentrate? | Density-based clustering | Descriptive | Hotspot clusters and noise | IV |
| Which combinations raise KSI risk? | Association rule mining | Descriptive | Rules with support, confidence, lift | V |
| Do crash types hide different patterns? | Categorical clustering, then rules per segment | Descriptive | Segments and their rules | IV, V |
| Is a collision likely to be KSI? | Binary classification | Predictive | Class and probability | III |
| Which sites will record KSI next year? | Classification and count prediction on grid cells | Predictive | Ranked sites | III |
| What should the authority do? | Decision support | Prescriptive use | Priority list | VI |

## 5. Suitable algorithms and their justification

### 5.1 DBSCAN for hotspots [8]

**How it works.** A point is a core point if at least `min_samples` points lie within radius `eps`. Clusters grow by joining reachable core points, and anything unreachable stays as noise. **Why here.** Most severe collisions are isolated, and a black spot is a small dense patch of any shape. K-Means would force every collision into a cluster, needs k in advance and assumes round clusters of similar size. Earlier hotspot work used kernel density with K-Means [12]; DBSCAN gives each hotspot an explicit boundary without choosing k. **Limit.** One `eps` cannot suit both central London and a rural road, and a large `eps` chains neighbouring junctions into one cluster, so settings are tested.

### 5.2 Apriori for association rules [9]

**How it works.** Each collision is a transaction of items such as *Speed: 60 mph*. Apriori uses the fact that every subset of a frequent itemset is frequent, which prunes the search. For a rule A → KSI:

`support = P(A and KSI)` `confidence = P(KSI | A)` `lift = P(KSI | A) / P(KSI)`

**Why here.** The attributes are categorical and the output is readable, so an engineer can act on it. It is also the method of the reference paper [4]. **Limit.** Rules show association, not cause.

### 5.3 K-Modes for segmentation [10]

**How it works.** Like K-Means, but it measures distance by the number of mismatched categories and represents each cluster by its most common values. **Why here.** Road accident data is mixed, and the reference paper reports that segmenting first reveals rules hidden in the whole data set [4]. Means have no meaning for categories such as lighting.

### 5.4 Decision Tree and Random Forest for classification [11]

**Decision Tree.** Splits greedily to reduce Gini impurity, `Gini(D) = 1 - sum(p_i^2)`. Trees have been used on accident severity before [13]. A pruned tree gives rules a traffic engineer can read and challenge. **Random Forest.** Averages many trees grown on bootstrap samples with random feature subsets, which reduces variance and captures interactions. It trades readability for accuracy, so both are used. Naive Bayes and AdaBoost are included as syllabus baselines.

| Algorithm | Role | Strength here | Weakness here |
|---|---|---|---|
| DBSCAN | Where | Finds any shape, keeps noise out, no k needed | Sensitive to `eps`; one density for all regions |
| Apriori | Why | Readable, exhaustive at a set support | Many redundant rules; association only |
| K-Modes | Segments | Works directly on categories | Result depends on k and starting points |
| Decision Tree | Explain | White-box rules | Unstable; lower accuracy than ensembles |
| Random Forest | Predict | Handles interactions, resists overfitting | Less transparent; heavier to train |

## 6. The data mining process




**Figure 4.** The eight-stage process applied to this case.

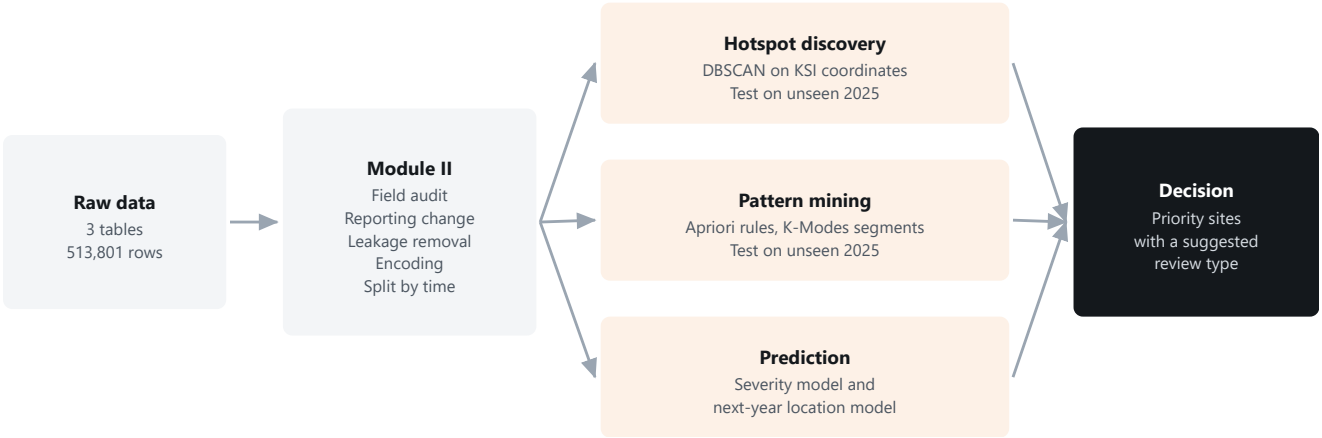



**Figure 5.** Technical architecture: one prepared data set feeds three branches, and all three feed the decision.

| Stage | In this case study |
|---|---|
| Problem | Decide where a limited safety budget covers the most KSI collisions |
| Data | Five years of STATS19 collisions, vehicles and casualties |
| Preprocessing | Audit, integrate, handle the reporting change, remove leakage, encode, split by time |
| Algorithm | DBSCAN, Apriori, K-Modes, Naive Bayes, Decision Tree, Random Forest, AdaBoost |
| Results | Hotspots, rules, segments, severity probabilities, site scores |
| Evaluation | Test everything on 2025, a year never used for mining or training |
| Insights | Name the places and combinations that matter, and state what the data cannot show |
| Decision | Ranked site list, each with a suggested review type for an engineer |

## 7. Expected and available results, and evaluation measures

### 7.1 Available results from the reference paper

On 11,574 Dehradun accidents from 2009 to 2014, Kumar and Toshniwal found six clusters [4]. They include two-wheeler accidents at intersections and curves (27.48%), two-wheeler accidents on highways near hills, forest and farmland (15.31%), falls from height (16.66%), multi-vehicle and fixed-object accidents with no light (12.04%), pedestrian hits (15.08%) and roll-overs (13.42%). Their main conclusion is that clustering before rule mining reveals information that rules from the entire data set do not.

### 7.2 Available results from our data exploration

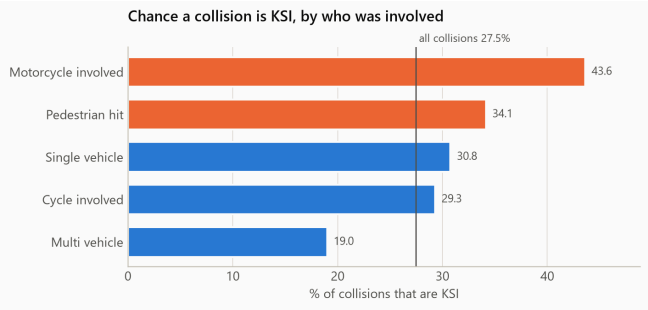



**Figure 6.** Motorcycle collisions are KSI 43.6% of the time, against 27.4% overall.

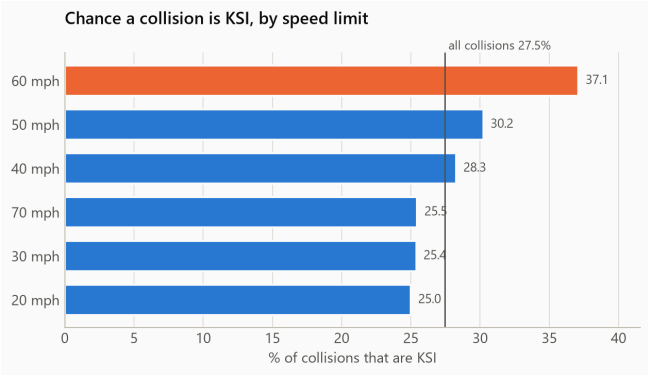



**Figure 7.** Severity peaks at 60 mph (37.1%), mostly rural single carriageways, not at 70 mph (25.5%), which is mostly motorway.

### 7.3 Evaluation measures

| Measure | What it tells us | Why it fits this case |
|---|---|---|
| **Recall**, TP / (TP + FN) | Share of KSI collisions the model catches | Missing a severe crash pattern costs most |
| **Precision**, TP / (TP + FP) | Share of flagged collisions that really are KSI | Engineers' time is limited |
| **F1** | Balance of precision and recall | Accuracy alone misleads on a minority class |
| **ROC-AUC** | Chance a KSI case is ranked above a non-KSI case | Ranking is the real use |
| **PR-AUC** | Precision across recall levels; chance level equals the KSI rate | More honest than ROC-AUC when positives are rarer |
| **Support, confidence, lift** | How common, how reliable and how far above base rate a rule is | Same measures as the reference paper |
| **Held-out rule lift** | Whether a rule still holds on 2025 | Rules that fail on new data are noise |
| **Hotspot coverage** | 2025 KSI collisions reached, in total and per $km^2$ | Shows whether hotspots persist |
| **Budget curve** | KSI collisions covered by the top N sites | This is the business success test |

### 7.4 Expected results

- **Hotspots:** most KSI collisions should remain noise, with a small share in street-scale clusters, mostly in large cities.
- **Rules:** vulnerable road users and high-speed rural roads should lead, matching the two-wheeler patterns in the reference paper.
- **Severity model:** moderate performance, well below the figures common in tutorials, because severity depends on factors never recorded, such as impact speed and seat belt use.
- **Location model:** better than a past-count list, with a modest gain, because past counts are already a strong signal.

Measured results for all of these are in the CA2 report.

## 8. How data mining supports the final decision

| Mining output | Decision it supports | Who acts |
|---|---|---|
| DBSCAN hotspots | Which places to inspect first | Road safety engineers |
| Rules by theme | What to check at a site: lighting, crossing, speed, junction layout | Engineers and police |
| Location model ranking | Next year's shortlist, and how many sites a budget covers | Budget holder |
| Severity model drivers | Which road users to target in campaigns | Road safety officers |

**What this analysis does not claim.** It finds associations, not causes, so it cannot promise that a treatment will prevent a crash. STATS19 has no traffic volumes, so a busy site scores high partly because it is busy. Only police-reported injury collisions are included. The output is a shortlist for engineers to investigate, not an automatic decision.

**Contribution to the goal.** The authority moves from reacting to last year's counts to a tested, repeatable process that says where to look, what to look for and how much of next year's severe crash burden a given budget covers. This supports SDG target 3.6 on road deaths and injuries [3].

## References


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